

Assignment 4: Modification of Python Wrapper Setup

In this assignment, a steady-state compressible turbulent flat plate simulation was performed using SU2. The objective was to modify the Python wrapper to implement a spatially varying wall temperature along the plate.

Methodology:

A spatially varying temperature profile was applied along the flat plate using the Python wrapper. The temperature at each node on the wall boundary was defined as a function of its x-coordinate.

The implemented temperature relation is:

$$T_{\text{wall}} = 300 + 500 * (x + 0.0175) / 0.035$$

This results in a temperature distribution that increases gradually along the length of the plate.

Configuration Details:

- Solver: RANS
- Turbulence Model: SST
- Flow Type: Steady-state
- Freestream Temperature: 293 K
- Boundary Conditions:
 - Plate: Python-controlled spatially varying temperature
 - Farfield: Standard farfield condition

The simulation was performed using the SU2 configuration file for flat plate flow.

Implementation:

The Python wrapper was modified to:

- Access coordinates of each node on the wall boundary
- Compute temperature based on x-coordinate
- Apply the computed temperature using custom boundary conditions

This allows precise control over the temperature distribution along the plate.

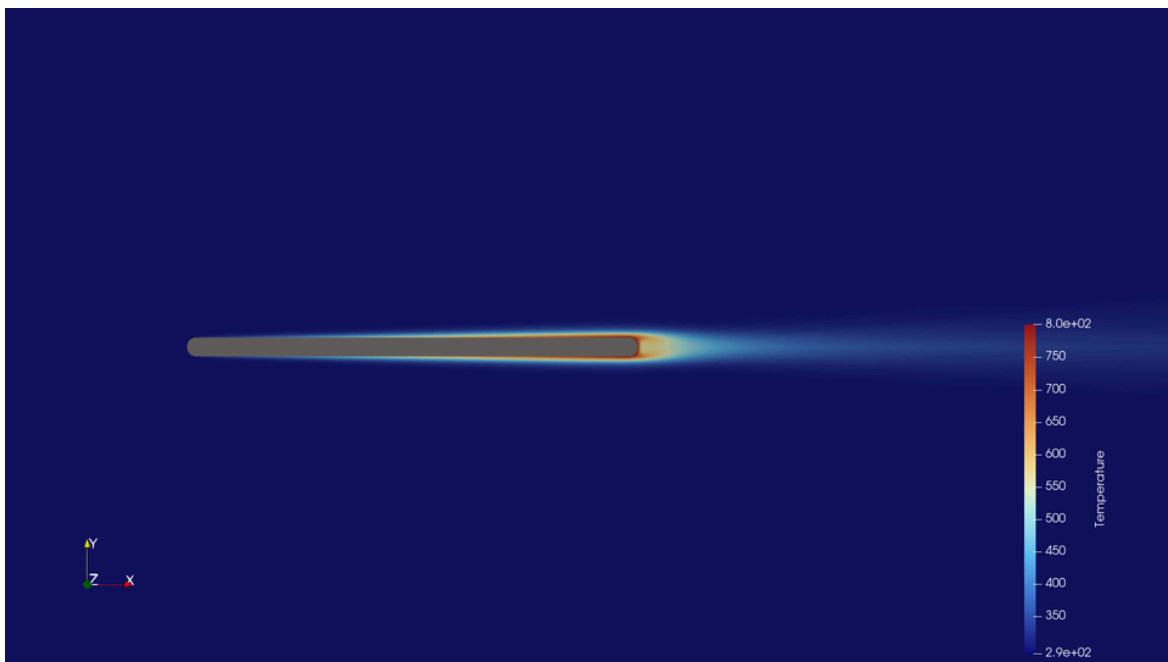
Results:

- The temperature contour shows a clear spatial variation along the flat plate.
- The temperature increases progressively from the leading edge to the trailing edge.
- The temperature varies approximately from 300 K near the leading edge to around 800 K near the trailing edge.
- A high-temperature region is observed near the downstream end of the plate, indicating successful implementation of the spatial temperature gradient. The surrounding flow also reflects thermal diffusion from the heated surface.

Conclusion:

The spatially varying wall temperature was successfully implemented using the SU2 Python wrapper. The results demonstrate the ability to apply customized boundary conditions and simulate realistic thermal gradients in CFD problems.

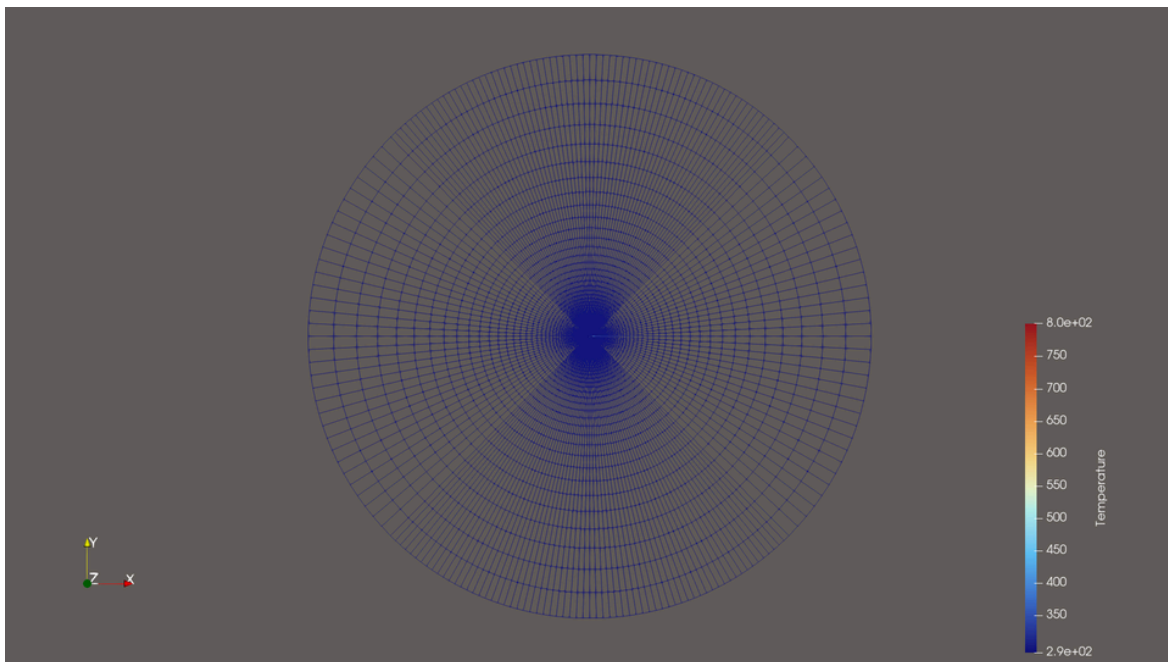
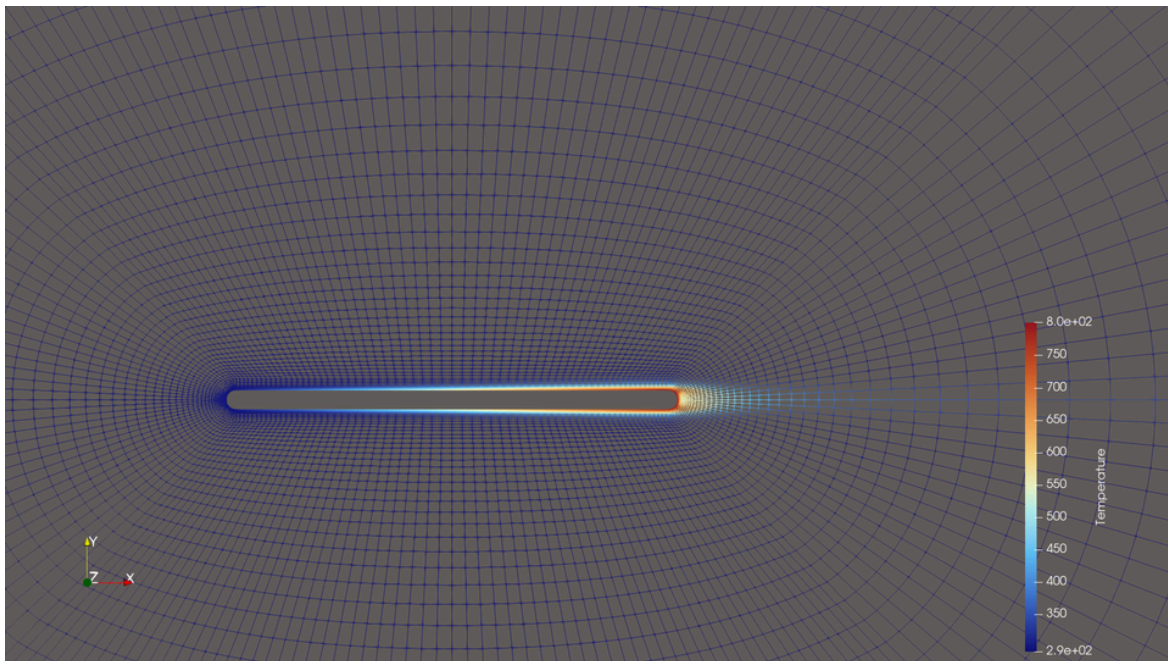
Outputs:



Surface View

Temperature contour showing spatial variation along the flat plate, with temperature increasing in the streamwise direction.

Wireframe View for Spatially Varying Temperature:





Feature Edge View highlighting the temperature variation along the flat plate boundary